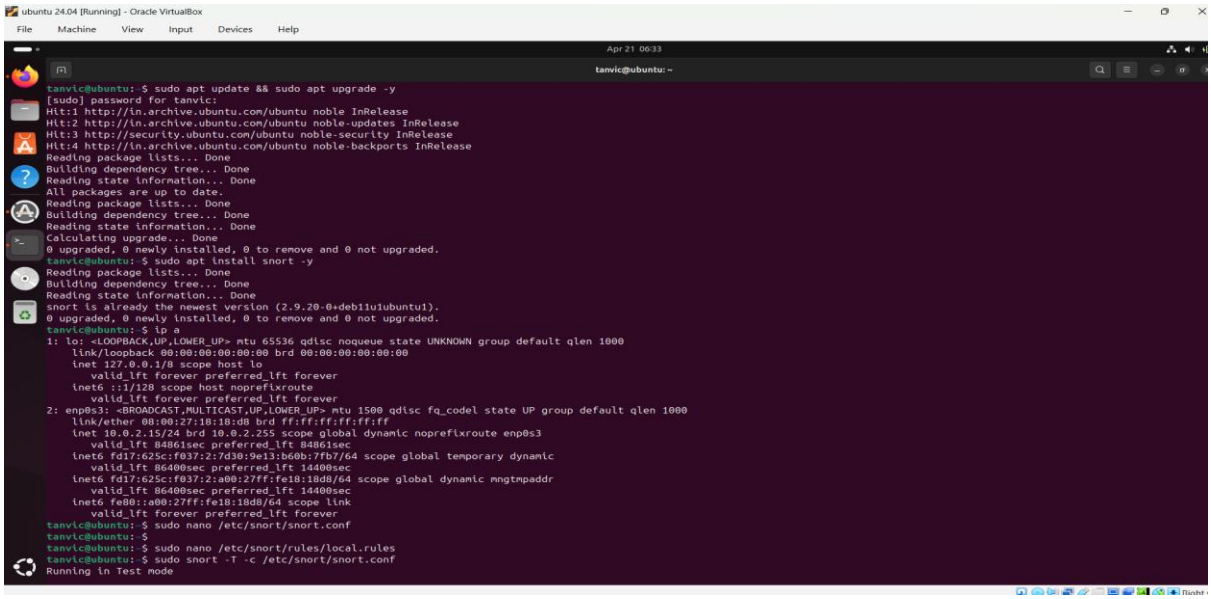
Firewall + IDS using iptables and Snort

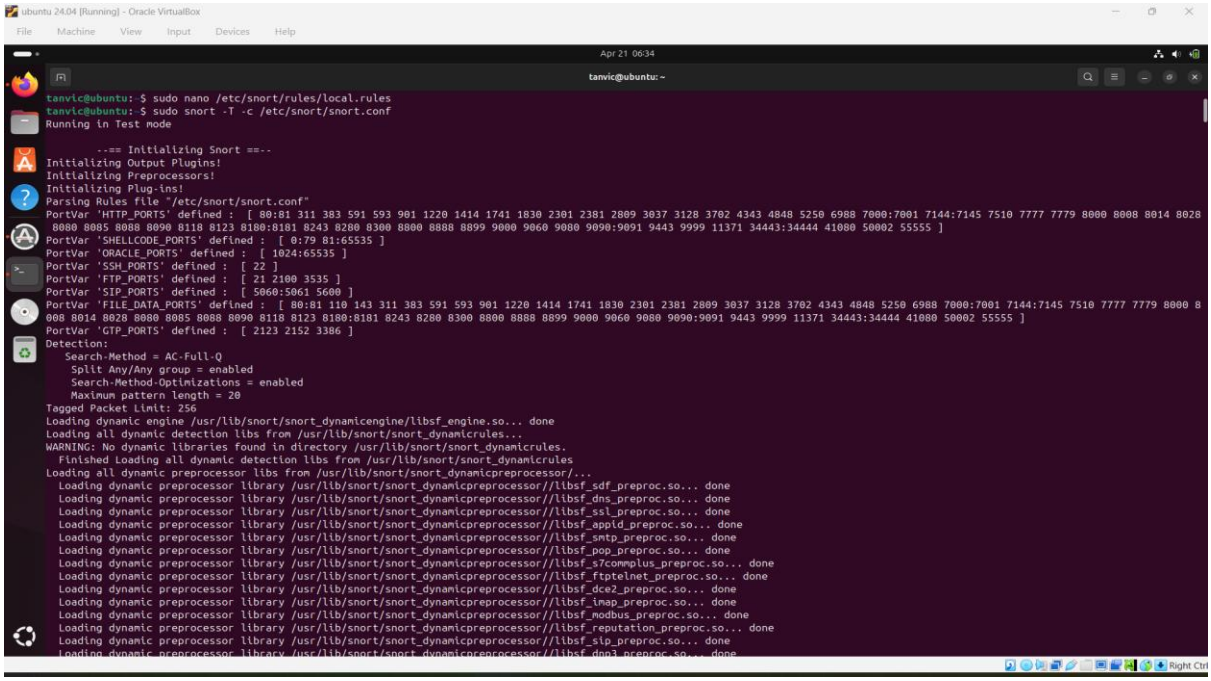
Requirements

You need:

- 2 systems (or VMs):
    - Attacker → Kali Linux
    - Target → Ubuntu/Linux

Use **VirtualBox** if doing on one laptop





```
        --== Initialization Complete ==--

   ,,_     -*> Snort! <*-
  o"  )~   Version 2.9.20 GRE (Build 82)
   ''''    By Martin Roesch & The Snort Team: http://www.snort.org/contact#team
           Copyright (C) 2014-2022 Cisco and/or its affiliates. All rights reserved.
           Copyright (C) 1998-2013 Sourcefire, Inc., et al.
           Using libpcap version 1.10.4 (with TPACKET_V3)
           Using PCRE version: 8.39 2016-06-14
           Using ZLIB version: 1.3

           Rules Engine: SF_SNORT_DETECTION_ENGINE  Version 3.2  <Build 1>
           Preprocessor Object: SF_SSH  Version 1.1  <Build 3>
           Preprocessor Object: SF_GTP  Version 1.1  <Build 1>
           Preprocessor Object: SF_DNP3  Version 1.1  <Build 1>
           Preprocessor Object: SF_SIP  Version 1.1  <Build 1>
           Preprocessor Object: SF_REPUTATION  Version 1.1  <Build 1>
           Preprocessor Object: SF_MODBUS  Version 1.1  <Build 1>
           Preprocessor Object: SF_IMAP  Version 1.0  <Build 1>
           Preprocessor Object: SF_DCERPC2  Version 1.0  <Build 3>
           Preprocessor Object: SF_FTPTELNET  Version 1.2  <Build 13>
           Preprocessor Object: SF_S7COMMPLUS  Version 1.0  <Build 1>
           Preprocessor Object: SF_POP  Version 1.0  <Build 1>
           Preprocessor Object: SF_SMTP  Version 1.1  <Build 9>
           Preprocessor Object: appid  Version 1.1  <Build 5>
           Preprocessor Object: SF_SSLPP  Version 1.1  <Build 4>
           Preprocessor Object: SF_DNS  Version 1.1  <Build 4>
           Preprocessor Object: SF_SDF  Version 1.1  <Build 1>
Commencing packet processing (pid=29454)
04/21-06:33:59.308566  [**] [1:1000001:0] Ping Detected [**] [Priority: 0] {IPV6-ICMP} fe80::a00:27ff:fe18:18d8 -> fe80::2
04/21-06:33:59.309033  [**] [1:1000001:0] Ping Detected [**] [Priority: 0] {IPV6-ICMP} fe80::2 -> fe80::a00:27ff:fe18:18d8
\04/21-06:37:23.770811  [**] [1:1000001:0] Ping Detected [**] [Priority: 0] {IPV6-ICMP} fe80::2 -> ff02::1
04/21-06:38:59.314093  [**] [1:1000001:0] Ping Detected [**] [Priority: 0] {IPV6-ICMP} fe80::a00:27ff:fe18:18d8 -> fe80::2
04/21-06:38:59.314976  [**] [1:1000001:0] Ping Detected [**] [Priority: 0] {IPV6-ICMP} fe80::2 -> fe80::a00:27ff:fe18:18d8
04/21-06:43:59.346081  [**] [1:1000001:0] Ping Detected [**] [Priority: 0] {IPV6-ICMP} fe80::a00:27ff:fe18:18d8 -> fe80::2
04/21-06:43:59.346726  [**] [1:1000001:0] Ping Detected [**] [Priority: 0] {IPV6-ICMP} fe80::2 -> fe80::a00:27ff:fe18:18d8
```

In New Terminal-

```
ubuntu 24.04 [Running] - Oracle VirtualBox
File  Machine  View  Input  Devices  Help
Apr 21 06:56
tanvic@ubuntu: ~
tanvic@ubuntu: ~
tanvic@ubuntu: ~
tanvic@ubuntu:~$ ping google.com
PING google.com (142.250.67.206) 56(84) bytes of data.
64 bytes from bom12s08-in-f14.1e100.net (142.250.67.206): icmp_seq=1 ttl=255 time=9.79 ms
64 bytes from bom12s08-in-f14.1e100.net (142.250.67.206): icmp_seq=2 ttl=255 time=6.98 ms
64 bytes from bom12s08-in-f14.1e100.net (142.250.67.206): icmp_seq=3 ttl=255 time=8.40 ms
64 bytes from bom12s08-in-f14.1e100.net (142.250.67.206): icmp_seq=4 ttl=255 time=8.29 ms
64 bytes from bom12s08-in-f14.1e100.net (142.250.67.206): icmp_seq=5 ttl=255 time=8.04 ms
64 bytes from bom12s08-in-f14.1e100.net (142.250.67.206): icmp_seq=6 ttl=255 time=7.90 ms
64 bytes from bom12s08-in-f14.1e100.net (142.250.67.206): icmp_seq=7 ttl=255 time=8.83 ms
```

```
Apr 21 06:58
tanvic@ubuntu: ~
tanvic@ubuntu: ~
tanvic@ubuntu: ~
    No of Allocs :            3
    No of Frees  :           48
  Total memory used :         1379

Heap Statistics of imap:
          Total Statistics:
               Memory in use:           1379 bytes
                No of allocs:              3
                 No of frees:             48
        Config Statistics:
               Memory in use:           1379 bytes
                No of allocs:              3
                 No of frees:             48
==============================================================================

Memory Statistics for File at:Tue Apr 21 06:58:07 2026

Total buffers allocated:          0
Total buffers freed:              0
Total buffers released:           0
Total file mempool:               0
Total allocated file mempool:     0
Total freed file mempool:         0
Total released file mempool:      0

Heap Statistics of file:
          Total Statistics:
               Memory in use:            280 bytes
                No of allocs:              6
                 No of frees:              1
        Session Statistics:
               Memory in use:              0 bytes
                No of allocs:              1
                 No of frees:              1
        Mempool Statistics:
               Memory in use:            280 bytes
                No of allocs:              5
                 No of frees:              0
==============================================================================
Snort exiting
tanvic@ubuntu:~$
Right Ctrl
Hot weather  Now
Search
ENG  IN
12:28 PM  21-04-2026
```

```
tanvic@ubuntu:~$ sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT
[sudo] password for tanvic:
tanvic@ubuntu:~$ sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT
tanvic@ubuntu:~$ sudo iptables -A INPUT -p tcp --dport 23 -j DROP
tanvic@ubuntu:~$ sudo iptables -A INPUT -j DROP
tanvic@ubuntu:~$ sudo iptables -L
sudo: unable to resolve host ubuntu: Temporary failure in name resolution
Chain INPUT (policy ACCEPT)
target     prot opt source               destination
ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:ssh
ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:http
DROP       tcp  --  anywhere             anywhere             tcp dpt:telnet
DROP       all  --  anywhere             anywhere

Chain FORWARD (policy ACCEPT)
target     prot opt source               destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination
tanvic@ubuntu:~$
```

```
Session  Actions  Edit  View  Help
┌──(kali㉿kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:01:09:07 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 80771sec preferred_lft 80771sec
    inet6 fd17:625c:f037:2:8105:cb88:ce89:ca41/64 scope global temporary dynamic
       valid_lft 86356sec preferred_lft 14356sec
    inet6 fd17:625c:f037:2:a00:27ff:fe01:907/64 scope global dynamic mngtmpaddr noprefixroute
       valid_lft 86356sec preferred_lft 14356sec
    inet6 fe80::a00:27ff:fe01:907/64 scope link noprefixroute
       valid_lft forever preferred_lft forever

┌──(kali㉿kali)-[~]
└─$ ping 10.0.2.15
PING 10.0.2.15 (10.0.2.15) 56(84) bytes of data.
64 bytes from 10.0.2.15: icmp_seq=1 ttl=64 time=0.040 ms
64 bytes from 10.0.2.15: icmp_seq=2 ttl=64 time=0.028 ms
64 bytes from 10.0.2.15: icmp_seq=3 ttl=64 time=0.030 ms
64 bytes from 10.0.2.15: icmp_seq=4 ttl=64 time=0.030 ms
64 bytes from 10.0.2.15: icmp_seq=5 ttl=64 time=0.030 ms
64 bytes from 10.0.2.15: icmp_seq=6 ttl=64 time=0.029 ms
64 bytes from 10.0.2.15: icmp_seq=7 ttl=64 time=0.048 ms
64 bytes from 10.0.2.15: icmp_seq=8 ttl=64 time=0.050 ms
64 bytes from 10.0.2.15: icmp_seq=9 ttl=64 time=0.047 ms
64 bytes from 10.0.2.15: icmp_seq=10 ttl=64 time=0.057 ms
64 bytes from 10.0.2.15: icmp_seq=11 ttl=64 time=0.056 ms
64 bytes from 10.0.2.15: icmp_seq=12 ttl=64 time=0.061 ms
64 bytes from 10.0.2.15: icmp_seq=13 ttl=64 time=0.039 ms
64 bytes from 10.0.2.15: icmp_seq=14 ttl=64 time=0.036 ms
64 bytes from 10.0.2.15: icmp_seq=15 ttl=64 time=0.042 ms
64 bytes from 10.0.2.15: icmp_seq=16 ttl=64 time=0.032 ms
64 bytes from 10.0.2.15: icmp_seq=17 ttl=64 time=0.028 ms
64 bytes from 10.0.2.15: icmp_seq=18 ttl=64 time=0.062 ms
64 bytes from 10.0.2.15: icmp_seq=19 ttl=64 time=0.042 ms
64 bytes from 10.0.2.15: icmp_seq=20 ttl=64 time=0.056 ms
64 bytes from 10.0.2.15: icmp_seq=21 ttl=64 time=0.058 ms
64 bytes from 10.0.2.15: icmp_seq=22 ttl=64 time=0.038 ms
64 bytes from 10.0.2.15: icmp_seq=23 ttl=64 time=0.041 ms
64 bytes from 10.0.2.15: icmp_seq=24 ttl=64 time=0.054 ms
64 bytes from 10.0.2.15: icmp_seq=25 ttl=64 time=0.057 ms
64 bytes from 10.0.2.15: icmp_seq=26 ttl=64 time=0.055 ms
64 bytes from 10.0.2.15: icmp_seq=27 ttl=64 time=0.046 ms
64 bytes from 10.0.2.15: icmp_seq=28 ttl=64 time=0.043 ms
64 bytes from 10.0.2.15: icmp_seq=29 ttl=64 time=0.042 ms
64 bytes from 10.0.2.15: icmp_seq=30 ttl=64 time=0.049 ms
64 bytes from 10.0.2.15: icmp_seq=31 ttl=64 time=0.041 ms
64 bytes from 10.0.2.15: icmp_seq=32 ttl=64 time=0.083 ms
64 bytes from 10.0.2.15: icmp_seq=33 ttl=64 time=0.042 ms
64 bytes from 10.0.2.15: icmp_seq=34 ttl=64 time=0.052 ms
64 bytes from 10.0.2.15: icmp_seq=35 ttl=64 time=0.058 ms
64 bytes from 10.0.2.15: icmp_seq=36 ttl=64 time=0.052 ms
```

```
64 bytes from 10.0.2.15: icmp_seq=169 ttl=64 time=0.024 ms
64 bytes from 10.0.2.15: icmp_seq=170 ttl=64 time=0.055 ms
64 bytes from 10.0.2.15: icmp_seq=171 ttl=64 time=0.051 ms
64 bytes from 10.0.2.15: icmp_seq=172 ttl=64 time=0.052 ms
64 bytes from 10.0.2.15: icmp_seq=173 ttl=64 time=0.050 ms
64 bytes from 10.0.2.15: icmp_seq=174 ttl=64 time=0.042 ms
64 bytes from 10.0.2.15: icmp_seq=175 ttl=64 time=0.050 ms
64 bytes from 10.0.2.15: icmp_seq=176 ttl=64 time=0.026 ms
64 bytes from 10.0.2.15: icmp_seq=177 ttl=64 time=0.024 ms
^C
--- 10.0.2.15 ping statistics ---
177 packets transmitted, 177 received, 0% packet loss, time 180052ms
rtt min/avg/max/mdev = 0.023/0.049/0.516/0.039 ms

┌──(kali㉿kali)-[~]
└─$ nmap 10.0.2.15
Starting Nmap 7.99 ( https://nmap.org ) at 2026-04-21 03:26 -0400
Nmap scan report for 10.0.2.15
Host is up (0.000056s latency).
Not shown: 999 filtered tcp ports (no-response)
PORT   STATE SERVICE
80/tcp open  http

Nmap done: 1 IP address (1 host up) scanned in 8.70 seconds

┌──(kali㉿kali)-[~]
```

```
┌──(kali㉿kali)-[~]
└─$ telnet 10.0.2.15
rying 10.0.2.15...
elnet: Unable to connect to remote host: Connection refused

┌──(kali㉿kali)-[~]
```

```
┌──(kali㉿kali)-[~]
└─$ sudo apt install iptables-persistent
[sudo] password for kali:
The following packages were automatically installed and are no longer required:
  curlftpfs                libcrypt-dev             libicu76                 libmupdf25.1             libpostproc58   libsphinxbase3t64   libzxing3                python3-aiocache         python3-yaswfp
  libaudio2                libdlt2                  libio-pty-perl           libonnx1t64              librubberband2  libswscale8         mesa-vdpau-drivers       python3-aiomcache        ruby-unf-ext
  libavfilter10            libdnnl3.6               libipc-run-perl          libonnxruntime-providers libsframe2      libteamdctl0        openjdk-21-jre           python3-fs               vdpau-driver-all
  libavformat61            libfuse2t64              libmjpegutils-2.1-0t64   libonnxruntime1.21       libsimdutf27    libtsk19t64         openjdk-21-jre-headless  python3-pysnmp4
  libblosc2-4              libgav1-1                libmpeg2encpp-2.1-0t64   libplacebo351            libsimdutf31    libvdpau-va-gl1     pocketsphinx-en-us       python3-terminaltables
  libconfig-inifiles-perl  libgdk-pixbuf2.0-bin     libmplex2-2.1-0t64       libpocketsphinx3         libsnmp40t64    libxnnpack0.20241108 postgresql-common-dev   python3-wapiti-arsenic
Use 'sudo apt autoremove' to remove them.

Installing:
  iptables-persistent

Installing dependencies:
  netfilter-persistent

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 124
  Download size: 18.7 kB
  Space needed: 96.3 kB / 12.2 GB available

Continue? [Y/n] y
Get:1 http://kali.download/kali kali-rolling/main amd64 netfilter-persistent all 1.0.24 [8,072 B]
Get:2 http://kali.download/kali kali-rolling/main amd64 iptables-persistent all 1.0.24 [10.7 kB]
Fetched 18.7 kB in 1s (26.2 kB/s)
Preconfiguring packages ...
Selecting previously unselected package netfilter-persistent.
(Reading database… 471875 files and directories currently installed.)
Preparing to unpack …/netfilter-persistent_1.0.24_all.deb…
Unpacking netfilter-persistent (1.0.24)…
Selecting previously unselected package iptables-persistent.
Preparing to unpack …/iptables-persistent_1.0.24_all.deb…
Unpacking iptables-persistent (1.0.24)…
Setting up netfilter-persistent (1.0.24)…
update-rc.d: We have no instructions for the netfilter-persistent init script.
update-rc.d: It looks like a non-network service, we enable it.
netfilter-persistent.service is a disabled or a static unit, not starting it.
Setting up iptables-persistent (1.0.24)…
Processing triggers for man-db (2.13.1-1)…
Processing triggers for kali-menu (2026.2.2)…
```

```
Session  Actions  Edit  View  Help
Package configuration

┌──────────────────────────────────────────────────────────────────────────────┐
│                        Configuring iptables-persistent                       │
│                                                                              │
│  Current iptables rules can be saved to the configuration file /etc/iptables/rules.v4. These rules will then be loaded automatically during system startup.
│                                                                              │
│  Rules are only saved automatically during package installation. See the manual page of iptables-save(8) for instructions on keeping the rules file up-to-date.
│                                                                              │
│  Save current IPv4 rules?                                                    │
│                                                                              │
│                          <Yes>                          <No>                 │
└──────────────────────────────────────────────────────────────────────────────┘
```

```
┌──(kali㉿kali)-[~]
└─$ sudo netfilter-persistent save
run-parts: executing /usr/share/netfilter-persistent/plugins.d/15-ip4tables save
run-parts: executing /usr/share/netfilter-persistent/plugins.d/25-ip6tables save

┌──(kali㉿kali)-[~]
```